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Open Geospatial Consortium

# OGC API - ENVIRONMENTAL DATA RETRIEVAL STANDARD - PART 3: PROFILES

STANDARD
Implementation

DRAFT

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# I ABSTRACT

The OGC API – Environmental Data Retrieval (EDR) Part 1: Core Standard was designed to be flexible and easy to understand and implement by Web developers. As OGC API-EDR has been widely adopted, communities of practice have identified the need to constrain certain aspects of this flexibility in order to improve interoperability between implementations within specific domains. These constraints are expressed through API-EDR Service Profiles, which define additional, domain-specific requirements on top of the Core Standard.

The OGC API-EDR Part 3: Service Profile Standard defines a consistent and interoperable approach for specifying such profiles. Its objective is to ensure that independent API-EDR implementations expose predictable and compatible behaviour when claiming conformance to the same service profile.

An OGC API-EDR Service Profile defines a constrained and well-specified set of requirements that an API-EDR service shall implement to support a clearly defined set of functionality. In particular, service profiles establish standardized rules for the definition and use of collections and when appropriate their instances, ensuring that services claiming conformance present a consistent interface to clients. Conformance to a service profile indicates that an implementation meets all requirements defined by that profile.

An OGC API-EDR implementation may support multiple service profiles. Each supported profile represents a distinct conformance target and imposes specific requirements on the service's behaviour and capabilities. Implementations shall advertise the service profiles they support through their conformance declaration, enabling clients to determine which profiles are available.

Each OGC API-EDR Service Profile consists of an OGC ModSpec compliant document that defines the service profile's compliance rules. These rules are classified as requirements (which are subject to testing) and recommendations or permissions (which are not tested). For every requirement defined in the document, a corresponding conformance test is provided to validate compliance with the service profile.

If a service profile introduces or restricts functionality that can be expressed using OpenAPI, it must also include an OpenAPI document that describes this functionality. A service profile OpenAPI definition, provides a precise, machine-readable description of the operations, parameters, and responses required by the profile. These OpenAPI definitions are intended solely to describe the functionality defined by the service profile and do not represent a complete service description or replace the primary OGC API-EDR OpenAPI definition.

Service profile OpenAPI definitions are distinct from the OpenAPI definition of an OGC API-EDR implementation. The implementation OpenAPI definition represents the full set of capabilities offered by the service, including functionality defined by one or more supported service profiles as well as any additional, non-profiled functionality. This separation allows service profiles to precisely define interoperable subsets of behaviour while enabling implementations to evolve and extend beyond the scope of individual profiles without compromising conformance.

# II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, ModSpec, API, OpenAPI, HTML, Profile

# PREFACE

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# IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

# V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- UK Met Office
- Heazeltech
- US NWS

# VI SUBMITTERS

All questions regarding this submission should be directed to the editor or the submitters:

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# VII CONTRIBUTORS

Additional contributors to this Standard include the following:

Individual name(s), Organization

# PREFACE

# PREFACE

**NOTE:** The aim of the OGC API-EDR Part 3: Service Profile standard is to ensure interoperability between EDR API implementations by defining a consistent approach to specifying a Profile of OGC API-EDR Part 1: Core. To achieve this, it is essential that service providers use a consistent approach when defining `collections` and `instances` of `collections`. An OGC API-EDR Profile will specify a set of Requirements that an EDR API implementation must support to be a profile-compliant implementation.

This standard specifies Requirements and Recommendations for a Profile definition, and also conforms to the OGC Mod Spec Standard.

It is envisaged that this approach may be useful for other OGC API Standards.

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# 1

# SCOPE

# 1 SCOPE

**NOTE:** This Standard defines how to specify a Service Profile for the OGC API-EDR Part 1: Core Standard. It only defines restrictive profiles, not profiles that extend the EDR API Standard with new functionality. An extension may not maintain backward compatibility with the EDR API. A service profile may specify requirements in addition to those defined in this standard, as long as those requirements comply with the OGC API-EDR Part 1: Core Standard.

Some parts of this Standard could be used by other OGC API Standards and implementations.

The restrictions are defined in requirements. The requirements are defined by using JSON Schema fragments, which can then be formally tested.

2

# CONFORMANCE

# 2 CONFORMANCE

Conformance to the OGC API-EDR-Part 3 Standard (this document) by a profile of the OGC API — Environmental Data Retrieval Standard can be tested by inspection. The test suite is provided in Annex A.

This Standard contains normative language and thus places requirements on conformance, or mechanism for adoption, of candidate standards to which this Standard applies. In particular:

- OGC API-EDR Requirements Class: Core specifies the core requirements which shall be met by all standards claiming conformance to this Standard.

3

# NORMATIVE REFERENCES

# 3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.


ISO: ISO 19106, *Geographic information — Profiles*. International Organization for Standardization, Geneva https://www.iso.org/standard/26011.html.

Mark Burgoyne, Dave Blodgett, Chuck Heazel, Chris Little: OGC 19-086r4, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2021). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.0.0.

https://docs.ogc.org/is/17-069r4/17-069r4.html, OGC API — Features — Part 1: Core, Open Geospatial Consortium (2022).

https://docs.ogc.org/is/19-072/19-072.html, OGC API - Common - Part 1: Core, Open Geospatial Consortium (2023).

http://docs.ogc.org/DRAFTS/20-024.html, OGC API - Common - Part 2: Geospatial Data (Draft), Open Geospatial Consortium

Policy SWG: OGC 08-131r3, *The Specification Model — Standard for Modular specifications*. Open Geospatial Consortium (2009). https://docs.ogc.org/pol/08-131r3/08-131r3.pdf.

OpenAPI Initiative (OAI). **OpenAPI Specification 3.0** [online]. 2024 [viewed 2025-01-03]. The latest patch version at the time of publication of this standard was 3.0.4, available at https://spec.openapis.org/oas/v3.0.4

OpenAPI Initiative (OAI). **OpenAPI Specification 3.1** [online]. 2024 [viewed 2025-01-03]. The latest patch version at the time of publication of this standard was 3.1.1, available at https://spec.openapis.org/oas/v3.1.1

4

# TERMS AND DEFINITIONS

# 4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

## 4.1. Collection

Body of resources that belong or are used together. An aggregate, set, or group of related resources.

[**SOURCE:** OGC 20-024]

## 4.2. Profile

An implementation case of a more general standard or set of standards.

## 4.3. Conformance Module; Conformance Test Module

A set of related conformance classes and their associated components.

**Note 1 to entry:** When no ambiguity is possible, the word `test` may be omitted. i.e. conformance test module is the same as conformance module. Conformance modules may be nested in a hierarchical way.

[**SOURCE:** OGC 08-131r5]

## 4.4. Conformance Class; Conformance Test Class

A set of conformance tests that must be passed to receive a single certificate of conformance.

**Note 1 to entry:** When no ambiguity is possible, the word *test* may be left out, so conformance test class maybe called a conformance class.

[**SOURCE:** OGC 08-131r5]

## 4.5. Conformance Test

A test, abstract or real, of one or more requirements contained within a standard, or set of standards.

[**SOURCE:** OGC 08-131r5]

## 4.6. Requirement

Expression in the content of a standard conveying criteria to be fulfilled if compliance with the standard is to be claimed and from which no deviation is permitted.

[**SOURCE:** OGC 08-131r5]

## 4.7. Requirements Class

An aggregate of requirements with a single standardization target type that must all be satisfied to pass a conformance test Class.

[**SOURCE:** OGC 08-131r5]

## 4.8. Requirements Module

A set of related requirements classes and their associated components.

[**SOURCE:** OGC 08-131r5]

## 4.9. Standardization Goal

A concise statement of the problem that the standard helps address and the strategy envisioned for achieving a solution. This strategy typically identifies real-world entities that need to be modified or constrained. At the abstract level, those entities are the Standardization Target Types.

[**SOURCE:** OGC 08-131r5]

## 4.10. Standardization Target

Entity to which some requirements of a standard apply.

**Note 1 to entry:** The standardization target is the entity which may receive a certificate of conformance for a requirements class.

[**SOURCE:** OGC 08-131r5]

## 4.11. Standardization Target Type

Type of entity or set of entities to which the requirement of a standard apply.

**Note 1 to entry:** For example, the standardization target type for The OGC API – Features Standard are Web APIs. The standardization target type for the CDB Standard is "datastore". It is important to understand that a standard's root standardization target type can have sub-types, and that there can be a hierarchy of target types. For example, a Web API can have sub types of client, server, security, and so forth. As such, each requirements class can have a standardization target type that is a sub-type of the root.

[**SOURCE:** OGC 08-131r5]

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# CONVENTIONS

# 5 CONVENTIONS

This sections provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

## 5.1. Identifiers

The normative provisions in this standard are denoted by the URI

`http://www.opengis.net/doc/spec/ogcapi-edr-3/1.0`

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

### 5.1.1. Shortcuts

In the interest of readability, the following terms will be used as shorthand for more complex text:

- Profile: A Profile is a standard or specification which restricts and/or extends an existing standard. This standard defines the rules for creating a profile of the OGC API — Environmental Data Retrieval Standard. The term “Profile” will be used in this document as shorthand for “profile of the OGC API — Environmental Data Retrieval Standard”.
- OGC API-EDR: The term OGC API-EDR will be used in this document as shorthand for the term “OGC API — Environmental Data Retrieval Standard”

6

# CONTEXT

# 6 CONTEXT

## 6.1. Standardization Goal

The goal of this Standard is to ensure interoperability between implementations of the OGC API — Environmental Data Retrieval Standard (OGC API-EDR).

The OGC API-EDR Standard does not try to address every possible application domain. Rather, it provides a foundation which can be tailored for a specific domain. The result of this tailoring is a domain specific "profile" of the EDR API Standard.

A significant risk to this approach is that, in the act of profiling, interoperability will be compromised. This risk can be mitigated by establishing rules for how the OGC API-EDR Standard can be profiled. The goal of this Standard is to define a set of rules sufficient to ensure interoperability while retaining the adaptability needed to support domain-specific requirements.

## 6.2. Standardization Target Type

The Standardization Target Type for this Standard is the set of standards and specifications which profile the OGC API — Environmental Data Retrieval Standard.

It is important to understand that:

- This Standard is a standard for writing standards. It does not address the EDR API implementation.
- This Standard is a profile of the OGC ModSpec Model — Part 1: Core — A Standard for Designing and Writing Modular Standards (ModSpec).
- Implementations of this Standard are Profiles of the OGC API — Environmental Data Retrieval Standard
- The profiling model used is defined in ISO 19106:2004 Geographic information — Profiles

7

# REQUIREMENTS CLASS CORE

# 7 REQUIREMENTS CLASS CORE

| REQUIREMENTS CLASS 1: REQUIREMENTS CLASS 'CORE' | |
|---|---|
| IDENTIFIER | `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| CONFORMANCE CLASS | Conformance class A.1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/conf-class-core` |
| TARGET-TYPE | OGC API-EDR Profile Standard |
| NORMATIVE STATEMENTS | Requirement 1: `/req/core/modspec`<br>Requirement 1-12: `/req/core/api`<br>Requirement 1-26: `/req/core/instanceid`<br>Requirement 2: `/req/core/edr-conformant`<br>Requirement 3: `/req/core/parameter-names`<br>Requirement 4: `/req/core/root`<br>Requirement 5: `/req/core/root-description`<br>Requirement 7: `/req/core/root-keywords`<br>Requirement 8: `/req/core/root-provider`<br>Requirement 9: `/req/core/root-contact`<br>Requirement 10: `/req/core/root-links`<br>Requirement 11: `/req/core/publishing`<br>Requirement 12: `/req/core/openapi`<br>Requirement 14: `/req/core/requirements-set`<br>Requirement 15: `/req/core/collectionid`<br>Requirement 16: `/req/core/extent`<br>Requirement 17: `/req/core/extent-spatial`<br>Requirement 18: `/req/core/extent-temporal`<br>Requirement 19: `/req/core/extent-vertical`<br>Requirement 20: `/req/core/extent-custom`<br>Requirement 21: `/req/core/data-query`<br>Requirement 22: `/req/core/output-format`<br>Requirement 23: `/req/core/data-query-position`<br>Requirement 24: `/req/core/data-query-radius`<br>Requirement 25: `/req/core/data-query-area`<br>Requirement 26: `/req/core/data-query-cube`<br>Requirement 27: `/req/core/data-query-trajectory`<br>Requirement 28: `/req/core/data-query-corridor`<br>Requirement 29: `/req/core/data-query-locations`<br>Requirement 30: `/req/core/data-query-instances`<br>Requirement 31: `/req/core/paging-support`<br>Requirement 32: `/req/core/status-codes`<br>Requirement 33: `/req/core/links`<br>Requirement 34: `/req/core/asynchronous` |

| REQUIREMENTS CLASS 1: REQUIREMENTS CLASS 'CORE' |
| --- |
| Requirement 35: `/req/core/pubsub` |

## 7.1. Profiling Requirements

The Profile is conformant with the ModSpec

| REQUIREMENT 1 | |
| --- | --- |
| **IDENTIFIER** | `/req/core/modspec` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A profile of the OGC API — Environmental Data Retrieval Standard *SHALL* be conformant to the OGC Modular Specification. |

Implementations of the Profile are conformant with OGC API-EDR Part 1: Core

| REQUIREMENT 2 | |
| --- | --- |
| **IDENTIFIER** | `/req/core/edr-conformant` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that a conformant implementation (standardization target) of that profile demonstrate conformance to the OGC API — Environmental Data Retrieval Standard. |

Profiles often focus on restricting the values of Path parameters. A Profile should clearly specify the requirements for these restrictions

## REQUIREMENT 3

| | |
|---|---|
| **IDENTIFIER** | `/req/core/parameter-names` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* the allowed values and definitions of `parameter_names`, then, |
| **A** | Requirements *SHALL* be defined which specify the `parameter_names` and their definitions. |
| **B** | The `parameter_names` requirement definitions *SHALL* specify the required `parameter_names` objects in full including:<br>• name,<br>• unit,<br>• data type and<br>• measurement duration |
| **STATEMENT** | for example:<br>`"parameter_names": {`<br>`"prmsl": {`<br>`"type": "Parameter",`<br>`"description": "Air pressure at sea level",`<br>`"unit": {`<br>`"label": "Pascals",`<br>`"symbol": {`<br>`"value": "Pa",`<br>`"type": "https://qudt.org/vocab/unit/PA"`<br>`}`<br>`},`<br>`"observedProperty": {`<br>`"id": "http://codes.wmo.int/grib2/codeflag/4.2/_0-3-1",`<br>`"label": "MSL Pressure"`<br>`}`<br>`},`<br>`"t2m": {`<br>`"type": "Parameter",`<br>`"description": "Air temperature at 2m",`<br>`"unit": {`<br>`"label": "Kelvin",`<br>`"symbol": {`<br>`"value": "K",`<br>`"type": "https://qudt.org/vocab/unit/K"`<br>`}`<br>`},`<br>`"observedProperty": {`<br>`"id": "http://codes.wmo.int/grib2/codeflag/4.2/0-0-0",`<br>`"label": "Air temperature at 2m"`<br>`}`<br>`},`<br>`"dd": {`<br>`"type": "Parameter",`<br>`"description": "Wind Direction",`<br>`"unit": {`<br>`"label": "degree true",`<br>`"symbol": {`<br>`"value": "°",`<br>`"type": "https://qudt.org/vocab/unit/DEG"`<br>`}` |

## REQUIREMENT 3

```
                    },
                    "observedProperty": {
                        "id": "http://codes.wmo.int/grib2/codeflag/4.2/0-2-0",
                        "label": "Wind Direction"
                    },
                    "measurementType": {
                        "method": "mean",
                        "duration": "-PT10M"
                    }
                },
                "ff": {
                    "type": "Parameter",
                    "description": "10m Wind Speed",
                    "unit": {
                        "label": "m/s",
                        "symbol": {
                          "value": "ms-1",
                          "type": "https://qudt.org/vocab/unit/M/s"
                        }
                    },
                    "observedProperty": {
                        "id": "http://codes.wmo.int/grib2/codeflag/4.2/0-2-1",
                        "label": "10m Wind Speed"
                    },
                    "measurementType": {
                        "method": "mean",
                        "duration": "-PT10M"
                    }
                }
            }
```

A Profile must be interoperable with other OGC API-EDR data providers. Any valid OGC API-EDR document should be valid under the profile. That means:

- If a data element is valid for OGC API-EDR, then it should not be prohibited under the profile
- Custom query elements defined by a Service Profile *SHALL* be optional, not mandatory, for data queries.
- It is valid for a profile to prohibit the production and population of OGC API-EDR optional elements by data providers within the profile's domain.

## 7.2. Platform Resources

OGC API - Common Standard defines a set of common capabilities which are applicable to any OGC Web API Standard and implementation instance. Those capabilities provide the platform upon which resource-specific APIs can be built. This section describes those capabilities and any modifications needed to better support spatio-temporal data resources.

**Table 1** — Platform Resource Paths

| PATH TEMPLATE | METHOD | RESOURCE |
|---|---|---|
| {root}/ | GET | Landing page |
| {root}/{api} | GET | API Description (optional) |
| {root}/conformance | GET | Conformance Classes |

Where:

- {root} = Base URI for the API server
- {api} = an identifier for an optional description of the API

### 7.2.1. API Landing Page

Path = {root}/

Dependencies

- OGC API — Common — Part 1: Core
- OGC API — Environmental Data Retrieval Standard

The landing page provides links that support exploration of the resources offered via the API. The most important component of a landing page is a list of links. The Landing Page resource is initially defined in the Core conformance class of the OGC API — Common — Part 1 Standard. The OGC API — Environmental Data Retrieval Standard does not make any changes to this definition. The normative JSON Schema for an EDR Landing Page is defined in the LandingPage.yaml document.

Profiles of the OGC API – Environmental Data Retrieval Standard are able to require a richer description of the API by mandating optional attributes. The requirements below define additional content rules that service profiles can enforce.

| PERMISSION 1 | |
|---|---|
| **IDENTIFIER** | `/per/core/multiple-collection-definition` |
| **STATEMENT** | A Profile MAY impose requirements on the complete OGC API service implementation or on a defined subset of the functionality it provides. |

| REQUIREMENT 4 | |
|---|---|
| **IDENTIFIER** | `/req/core/root` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | While the OGC API-EDR Landing Page schema requires only the links property, a Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require the following additional properties and content: |
| **A** | The `title` property *SHALL* be required and populated |
| **B** | The `links` property *SHALL* define the links that SHALL be included in the Root response. |
| **C** | Each `link` defined in the `links` property *SHALL* be populated with href and rel properties. |

## RECOMMENDATION 1

| | |
|---|---|
| **IDENTIFIER** | `/rec/core/root` |
| **STATEMENT** | While the OGC API-EDR Landing Page schema requires only the links property, a Profile of the OGC API – Environmental Data Retrieval Standard *SHOULD* mandate the following additional properties. |
| **A** | The `description` property *SHOULD* be required |
| **B** | The `keywords` property *SHOULD* be required |
| **C** | The `provider` property *SHOULD* be required and populated with the `name` and `url` properties |
| **D** | The `contact` property *SHOULD* be required and populated with the with the `email` properties |

## REQUIREMENT 5

| | |
|---|---|
| **IDENTIFIER** | `/req/core/root-description` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that when an EDR Landing Page includes the `description` property, that property *SHALL* be populated. |

## REQUIREMENT 6

| | |
|---|---|
| **IDENTIFIER** | `/req/core/root-attribution` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that when an EDR Landing Page includes the `attribution` property, that property *SHALL* be populated. |

## REQUIREMENT 7

| | |
|---|---|
| **IDENTIFIER** | `/req/core/root-keywords` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that when an EDR Landing Page includes the `keywords` property, that property *SHALL* be populated with at least one keyword entry. |

## REQUIREMENT 8

| | |
|---|---|
| **IDENTIFIER** | `/req/core/root-provider` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that when an EDR Landing Page includes the `provider` property. |
| **A** | A `provider` property *SHALL* be populated with a `name` property. |

## PERMISSION 2

| | |
|---|---|
| **IDENTIFIER** | `/per/core/provider-url-definition` |
| **STATEMENT** | A `provider` property *MAY* be populated with a `url` property. |

## REQUIREMENT 9

| | |
|---|---|
| **IDENTIFIER** | `/req/core/root-contact` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that when an EDR Landing Page includes the `contact` property, |
| **A** | A `contact` property *SHALL* be populated with the `email` property. |

## PERMISSION 3

| | |
|---|---|
| **IDENTIFIER** | `/per/core/contact-name-definition` |
| **STATEMENT** | A `contact` property *MAY* be populated with the `name` property. |

## PERMISSION 4

| | |
|---|---|
| **IDENTIFIER** | `/per/core/contact-url-definition` |
| **STATEMENT** | A `contact` property *MAY* be populated with the `url` property. |

## REQUIREMENT 10

| | |
|---|---|
| **IDENTIFIER** | `/req/core/root-links` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that when an EDR Landing Page includes the `links` property: |
| **A** | The `links` property *SHALL* define the links that *SHALL* be included in the Root response |
| **B** | The `links` property *SHALL* be populated with `href` and `rel` properties |

## PERMISSION 5

| | |
|---|---|
| **IDENTIFIER** | `/per/core/links-type-href-length-definition` |
| **STATEMENT** | The `links` property *MAY* be populated with type, hreflang and length properties |

## RECOMMENDATION 2

| | |
|---|---|
| **IDENTIFIER** | `/rec/core/root-links` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHOULD* require that when an EDR Landing Page includes the `links` property, the title property of each link *SHALL* be populated. |

### 7.2.2. API Definition

Path = {root}/{api}

Dependencies

- OGC API — Common — Part 1: Core
- OGC API — Environmental Data Retrieval Standard

Every OGC API is required to provide a definition document that describes the capabilities of that API. This definition document can be used by developers to understand the API, by software clients to connect to the server, or by development tools to support the implementation of servers and clients. The API Definition resource is initially defined in the Core conformance class of the OGC API — Common — Part 1 Standard. The OGC API — Environmental Data Retrieval Standard does not make any changes to that definition.

Each service profiles of the OGC API – Environmental Data Retrieval standard must provide an OpenAPI 3.1 document when the profile includes requirements that can be expressed using the OpenAPI specification. A service profile document is a subset of the API definition produced by the OGC API-EDR implementation and reflects the additional requirements specified by the profile.

**NOTE:** At this time only OpenAPI 3.0 and OpenAPI 3.1 documents are supported by OGC Web API Standards.

| REQUIREMENT 11 | |
|---|---|
| **IDENTIFIER** | `/req/core/publishing` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | An EDR service profile *SHALL* be published as an OpenAPI JSON document when it contains requirements that can be encoded in the OpenAPI specification. |
| **A** | The OpenAPI compatible rules described in the requirements *SHALL* be encoded using the OpenAPI 3.1 specification. |
| **B** | The OpenAPI compatible requirement rules *SHALL* be encoded in either the OpenAPI Path Item or in the Response object schema sections of the document. |

| REQUIREMENT 12 | |
|---|---|
| **IDENTIFIER** | `/req/core/openapi` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | A profile OpenAPI document *SHALL* describe the profile EDR API as follows: |
| **A** | The servers attributes of the OpenAPI root object *SHALL* be blank (the profile is not linked to specific implementations) |
| **B** | The Extent requirement rules *SHALL* be encoded in the JSON schema defined in the 200 responses for the /collections and /collections/{collectionId} Paths object |
| **C** | data_query requirement rules *SHALL* be encoded in the JSON schema defined in the 200 responses for the /collections and /collections/{collectionId} Paths object |
| **D** | The conformance classes *SHALL* be encoded in the JSON schema defined in the 200 responses for the /conformance Paths object. |
| **E** | The `data_query` types *SHALL* be encoded as Paths objects in the OpenAPI document, where appropriate the `output_format`, `default_output_format`, `crs`, `within_units`, `width-units`, `height-units` and `limit` (paging) requirements SHALL be encoded in the child Parameter objects of the Paths object. |
| **F** | The output_format requirement rules *SHALL* be encoded in the 200 responses of the data_query type Paths objects |
| **G** | The `parameter_names` requirements *SHALL* be encoded in the JSON schema defined in the 200 responses for the /collections and /collections/{collectionId} Paths object. |
| **H** | Where a `Collection` has `Instances`, `parameter_names` requirements *SHALL* be encoded in the JSON schema defined in the 200 responses for the /collections/{collectionId}/instances and /collections/{collectionId}/instances/{instanceId} Paths object. |
| **I** | An EDR API *SHALL* advertise the location of the profile OpenAPI document it complies with in the `links` section of the API root with a link relation value of `profile` |

### 7.2.3. Declaration of Conformance Classes

Path = {root}/conformance

Dependencies

- OGC API — Common — Part 1: Core
- OGC API — Environmental Data Retrieval Standard

To support "generic" clients that want to access implementations of multiple OGC API Standards and extensions — and not "just" a specific API server, the API has to declare the conformance classes it claims to have implemented. The Conformance Classes resource is initially defined in the Core conformance class of the OGC API — Common — Part 1 Standard. The OGC API — Environmental Data Retrieval Standard does not make any changes to this definition.

Profiles of the OGC API — Environmental Data Retrieval Standard have additional requirements governing which Conformance Classes and identifiers must be included in this resource.

| REQUIREMENT 13 | |
|---|---|
| **IDENTIFIER** | `/req/core/openapi-version` |
| **STATEMENT** | A Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* specify the versions of OpenAPI that an implementation *SHALL* support. |

**NOTE 1:** OpenAPI 3.0 and OpenAPI 3.1 are two distinct Conformance Classes in the OGC API-EDR Standard. This requirement can be addressed in a Profile by including the appropriate conformance classes at {root}/conformance.

**NOTE 2:** Profile developers should get guidance from the OGC Naming Authority on valid URIs for Profiles.

## 7.3. Spatio-temporal and Information Resources

**Table 2** — Spatio-temporal and Information Resource Paths

| PATH TEMPLATE | METHOD | RESOURCE |
|---|---|---|
| {root}/collections | GET | Metadata describing the Collections of data available from this API. |
| {root}/collections/{collectionId} | GET | Metadata describing the `Collection` of data which has the unique identifier {collectionId} |

Where:

- {root} = Base URI for the API server
- {collectionId} = an identifier for a specific `Collection` of data

### 7.3.1. Collections

OGC API implementations typically organize their geospatial resources into collections. Information about those Collections is accessed through the /collections path and the http://www.opengis.net/def/rel/ogc/1.0/data link relation.

Path = {root}/collections

Dependencies

- OGC API — Common — Part 2: Geospatial Data
- OGC API — Environmental Data Retrieval Standard

The Collections resource is initially defined in the `Collections` conformance class of the OGC API — Common — Part 2 Standard. The OGC API — Environmental Data Retrieval Standard does not make any changes to that definition.

An implementation of the OGC API-EDR may support multiple collections. Additional requirements address how the Service Profile should document requirements at the per-collection level as well as on the landing page (where appropriate)

**NOTE:** A Service profile can apply to multiple `Collections` and an OGC API EDR implementation can support multiple `Service Profiles`.

| REQUIREMENT 14 | |
|---|---|
| IDENTIFIER | `/req/core/requirements-set` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| A | The collection SHALL clearly identify the service profile that applies. |
| B | The collection SHOULD provide a link to the service profile definition with a link relation type of `profile`. |
| C | The service profile SHALL consist of a set of requirements for a `Collection` and (if the `Collection` supports `Instances`) the `Instances` of the `Collection`. For each of the attributes listed, if it is in the `Collection` (or instance), there SHALL be a requirement to define it. |

| PERMISSION 6 | |
|---|---|
| IDENTIFIER | `/per/core/multiple-service-profile-definition` |
| STATEMENT | An OGC API EDR implementation MAY include multiple service profiles. |

### 7.3.2. Collection Description

Each resource `Collection` is described by a set of metadata. That metadata can be accessed directly using the /collections/{collectionId} path and as an entry in the Collections property of the /collections response.

Path:

- {root}/collections (returns metadata for every `Collection`)
- {root}/collections/{collectionId} (returns metadata for the specified `Collection`)

Dependencies

- OGC API — Common — Part 2: Geospatial Data
- OGC API — Environmental Data Retrieval Standard

#### 7.3.2.1. Collection ID parameter restrictions

| REQUIREMENT 15 | |
|---|---|
| IDENTIFIER | `/req/core/collectionid` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* the valid values of the `collectionId` parameter, then: |
| A | The Service Profile *SHALL* specify the rules that the `collectionId` values must follow. |
| B | Those rules *SHALL* include a brief description explaining how the `collectionId` is generated. |
| C | Those rules *SHALL* be specified using either:<br>• identifier string or<br>• Regular expression defining valid string patterns. |

#### 7.3.2.2. Extent property restrictions

The `Collection` metadata includes an Extent property which defines a spatio-temporal envelope that encompasses the geospatial data in the `Collection`.

| REQUIREMENT 16 | |
|---|---|
| IDENTIFIER | `/req/core/extent` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | A Service Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* define a requirement defining a `bbox` which specifies the minimum spatial bounds that *SHALL* be supported |

| RECOMMENDATION 3 | |
|---|---|
| IDENTIFIER | `/rec/core/extent` |
| STATEMENT | A requirement *SHOULD* be defined specifying the rules for defining the `Collection` extent. |

## REQUIREMENT 17

| | |
|---|---|
| **IDENTIFIER** | `/req/core/extent-spatial` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **NOTE** | Regular expressions could be used to restrict reference system definitions to WKT2 or EPSG values |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard supports Extents with spatial dimensions, then: |
| **A** | The Service Profile *SHALL* specify the rules for the spatial Coordinate Reference System (CRS). |
| **B** | Those rules *SHALL* be specified using either:<br>• Enumerated list of valid CRS values<br>• Regular expression defining valid CRS string patterns. |

## PERMISSION 7

| | |
|---|---|
| **IDENTIFIER** | `/per/core/extent-spatial` |
| **STATEMENT** | Regular expressions *MAY* be used to restrict reference system definitions to WKT2 or EPSG values |

## REQUIREMENT 18

| | |
|---|---|
| **IDENTIFIER** | `/req/core/extent-temporal` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard supports Extents with a temporal dimension, then: |
| **A** | The Service Profile *SHALL* specify the rules for expressing the Temporal Reference System (TRS). |
| **B** | Those rules *SHALL* be specified using either:<br>• Enumerated list of valid TRS values<br>• Regular expression defining valid TRS string patterns. |

## RECOMMENDATION 4

| | |
|---|---|
| **IDENTIFIER** | `/rec/core/extent-temporal` |
| **STATEMENT** | A requirement *SHOULD* be defined specifying the minimum temporal bounds that *SHALL* be supported |

## REQUIREMENT 19

| | |
|---|---|
| **IDENTIFIER** | `/req/core/extent-vertical` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard supports Extents with a vertical dimension, then: |
| **A** | The Service Profile *SHALL* specify the rules for expressing the Vertical Reference System (VRS). |
| **B** | Those rules *SHALL* be specified using either:<br>• Enumerated list of valid VRS values<br>• Regular expression defining valid VRS string patterns. |

## RECOMMENDATION 5

| | |
|---|---|
| **IDENTIFIER** | `/rec/core/extent-vertical` |
| **STATEMENT** | A requirement *SHOULD* be defined specifying the minimum vertical bounds that *SHALL* be supported |

## REQUIREMENT 20

| | |
|---|---|
| **IDENTIFIER** | `/req/core/extent-custom` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard supports Extents with a custom dimension, then: |
| **A** | The Service Profile *SHALL* specify the rules for expressing the custom dimension. |
| **B** | Those rules *SHALL* be specified using either:<br>• A custom dimension id<br>• A custom dimension reference value<br>• A custom interval value<br>• An enumerated list of valid custom dimension values |

## RECOMMENDATION 6

| | |
|---|---|
| **IDENTIFIER** | `/rec/core/extent-custom` |
| **STATEMENT** | A requirement *SHOULD* be defined specifying the minimum bounds of custom extents that *SHALL* be supported |

## 7.4. Query Resources

**Table 3** — Query Resource Paths

| PATH TEMPLATE | METHOD | RESOURCE |
|---|---|---|
| {root}/collections/{collectionId}/{queryType} | GET, POST (Optional) | Retrieve data according to the query pattern from a `Collection` with the unique identifier {collectionId} |
| {root}/collections/{collectionId}/instances | GET | Retrieve metadata about Instances of a `Collection` |
| {root}/collections/{collectionId}/instances/{instanceId} | GET | Retrieve metadata from a specific `Instance` of a `Collection` with the unique identifiers{collectionId} and {instanceId} |
| {root}/collections/{collectionId}/instances/{instanceId}/{queryType} | GET, POST (Optional) | Retrieve data according to the query pattern from a specific `Instance` of a `Collection` with the unique identifiers {collectionId} and {instanceId} |

Where:

- {root} = Base URI for the API server
- {collectionId} = an identifier for a specific `Collection` of data
- {instanceId} = an identifier for a specific version or `Instance` of a `Collection` of data
- {queryType} = an identifier for a specific query pattern to retrieve data from a specific `Collection` of data

Path = {root}/collections/{collectionId}/{queryType}

Dependencies: OGC API — Environmental Data Retrieval Standard Part 1: Core

| REQUIREMENT 21 | |
|---|---|
| IDENTIFIER | `/req/core/data-query` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | A Service Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require definition of the supported data queries. |
| A | The data_queries definitions *SHALL* specify which data queries a collection supports. This can be defined as follows:<br>• Enumerated list of query types |
| B | Each data_query type listed *SHALL* have a requirement definition. |

## 7.4.1. Parameters

The following parameters are supported by all OGC API-EDR queries.

### 7.4.1.1. Output Format parameter

Also known as the -f parameter.

Data format for the output data (available options are listed in the Collections response).

| REQUIREMENT 22 | |
|---|---|
| IDENTIFIER | `/req/core/output-format` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | For every output_format specified in any of the data_query enumerated lists, there *SHALL* be a requirement which defines the schema or structure of the data (depending on the format). |

| RECOMMENDATION 7 | |
|---|---|
| IDENTIFIER | `/rec/core/output-format` |
| STATEMENT | The recommended definition approaches are as follows:<br>• JSON – Link to a JSON Schema definition<br>• XML – Link to a XML Schema definition<br>• CSV, TSV, PSV, SSV – Link to a definition based on the CSV on the web recommendations available from the CSV on the Web Working Group.<br>• Other types (e.g. binary file types) – Link to a description of the format |

## 7.4.2. Position Query

The Position query returns data for the requested coordinate. Logic for identifying the best match for the coordinate will depend on the `Collection` and is at the discretion of the query service implementer.

Path = {root}/collections/{collectionId}/position

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 23 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-position` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `position` query mandatory, then: |
| A | The Service Profile *SHALL* include a requirement mandating the `position` query. |
| B | The `Position` query requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of the operations that the query supports (i.e. GET, POST) |
| C | The `position` query requirement *SHALL* also specify the logic used in selecting the data returned by the response, i.e. exact, nearest neighbour, most representative or interpolated. |

## PERMISSION 8

| | |
|---|---|
| **IDENTIFIER** | `/per/core/position-mandatory-query-definition` |
| **STATEMENT** | The `position` query requirement *MAY* specify which OGC API-EDR `position` query parameters are mandatory for compliance with the profile. |

## PERMISSION 9

| | |
|---|---|
| **IDENTIFIER** | `/per/core/position-timestep-diff-query-definition` |
| **STATEMENT** | The `position` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

## PERMISSION 10

| | |
|---|---|
| **IDENTIFIER** | `/per/core/position-vertical-diff-query-definition` |
| **STATEMENT** | The `position` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple vertical levels, but not all parameters have data for the requested levels. |

## PERMISSION 11

| | |
|---|---|
| **IDENTIFIER** | `/per/core/position-custom-dimension-diff-query-definition` |
| **STATEMENT** | The `position` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

## PERMISSION 12

| | |
|---|---|
| **IDENTIFIER** | `/per/core/position-choice-query-definition` |
| **STATEMENT** | The `position` query requirement *MAY* specify rules on how the service should identify the most appropriate value for the requested position if there is no exact match (i.e. nearest, most representative, interpolated or none). |

### 7.4.3. Radius Query

The Radius query returns data within the defined radius of the requested coordinate.

Path = {root}/collections/{collectionId}/radius

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 24 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-radius` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `Radius` query mandatory, then: |
| A | The Service Profile *SHALL* include a requirement mandating the `Radius` query. |
| B | The `radius` query requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of `within_units` values<br>• Enumerated list of the operations that the query supports (i.e. GET, POST) |

| PERMISSION 13 | |
|---|---|
| IDENTIFIER | `/per/core/radius-mandatory-query-definition` |
| STATEMENT | The `radius` query requirement *MAY* specify which OGC API-EDR `radius` query parameters are mandatory for compliance with the profile. |

| PERMISSION 14 | |
|---|---|
| IDENTIFIER | `/per/core/radius-timestep-diff-query-definition` |
| STATEMENT | The `radius` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

| PERMISSION 15 | |
|---|---|
| IDENTIFIER | `/per/core/radius-vertical-diff-query-definition` |
| STATEMENT | The `radius` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple vertical levels, but not all parameters have data for the requested levels. |

| PERMISSION 16 | |
|---|---|
| IDENTIFIER | `/per/core/radius-custom-dimension-diff-query-definition` |
| STATEMENT | The `radius` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

### 7.4.4. Area Query

The Area query returns data within the polygon defined by the coords parameter. Logic for identifying the best match for the requested area will depend on the Collection and is at the discretion of the query service implementer.

Path = {root}/collections/{collectionId}/area

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 25 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-area` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `area` query mandatory, then: |
| A | The Service Profile *SHALL* include a requirement mandating the `area` query. |
| B | The `area` query requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of the operations that the query supports (i.e. GET, POST) |

| PERMISSION 17 | |
|---|---|
| IDENTIFIER | `/per/core/area-mandatory-query-definition` |
| STATEMENT | The `area` query requirement *MAY* specify which OGC API-EDR `area` query parameters are mandatory for compliance with the profile. |

| PERMISSION 18 | |
|---|---|
| IDENTIFIER | `/per/core/area-timestep-diff-query-definition` |
| STATEMENT | The `area` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

| PERMISSION 19 | |
|---|---|
| IDENTIFIER | `/per/core/area-vertical-diff-query-definition` |
| STATEMENT | The `area` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple vertical levels, but not all parameters have data for the requested levels. |

| PERMISSION 20 | |
|---|---|
| IDENTIFIER | `/per/core/area-custom-dimension-diff-query-definition` |
| STATEMENT | The `area` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

### 7.4.5. Cube Query

The Cube query returns a data cube defined by the bbox and z parameters.

Path = {root}/collections/{collectionId}/cube

Dependencies: OGC API — Environmental Data Retrieval Standard

## REQUIREMENT 26

| | |
|---|---|
| **IDENTIFIER** | `/req/core/data-query-cube` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `cube` query mandatory, then: |
| **A** | The Service Profile *SHALL* include a requirement mandating the `cube` query. |
| **B** | The `Cube` query requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of the operations that the query supports (i.e. GET, POST) |

## PERMISSION 21

| | |
|---|---|
| **IDENTIFIER** | `/per/core/cube-mandatory-query-definition` |
| **STATEMENT** | The `cube` requirement *MAY* specify which OGC API-EDR `cube` query parameters are mandatory for compliance with the profile. |

## PERMISSION 22

| | |
|---|---|
| **IDENTIFIER** | `/per/core/cube-timestep-diff-query-definition` |
| **STATEMENT** | The `cube` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

## PERMISSION 23

| | |
|---|---|
| **IDENTIFIER** | `/per/core/cube-vertical-diff-query-definition` |
| **STATEMENT** | The `cube` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple vertical levels, but not all parameters have data for the requested levels. |

| PERMISSION 24 | |
|---|---|
| IDENTIFIER | `/per/core/cube-resolution-x` |
| STATEMENT | The `cube` query requirement *MAY* specify rules that constrain the valid range of resolution-x values. |

| PERMISSION 25 | |
|---|---|
| IDENTIFIER | `/per/core/cube-resolution-y` |
| STATEMENT | The `cube` query requirement *MAY* specify rules that constrain the valid range of resolution-y values. |

| PERMISSION 26 | |
|---|---|
| IDENTIFIER | `/per/core/cube-resolution-z` |
| STATEMENT | The `cube` query requirement *MAY* specify rules that constrain the valid range of resolution-z values. |

| PERMISSION 27 | |
|---|---|
| IDENTIFIER | `/per/core/cube-custom-dimension-diff-query-definition` |
| STATEMENT | The `cube` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

### 7.4.6. Trajectory Query

The Trajectory query returns data along the path defined by the coords parameter. Logic for identifying the best matches for the requested trajectory will depend on the `Collection` and is at the discretion of the query service implementer.

Path = {root}/collections/{collectionId}/trajectory

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 27 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-trajectory` |

## REQUIREMENT 27

| | |
|---|---|
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `trajectory` query mandatory, then: |
| **A** | The Service Profile *SHALL* include a requirement mandating the `Ttrajectory` query. |
| **B** | The `trajectory` query requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of the operations that the query supports (i.e. `GET`, `POST`) |

## PERMISSION 28

| | |
|---|---|
| **IDENTIFIER** | `/per/core/trajectory-mandatory-query-definition` |
| **STATEMENT** | The `trajectory` requirement *MAY* specify which OGC API-EDR `trajectory` query parameters are mandatory for compliance with the profile. |

## PERMISSION 29

| | |
|---|---|
| **IDENTIFIER** | `/per/core/trajectory-timestep-diff-query-definition` |
| **STATEMENT** | The `trajectory` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

## PERMISSION 30

| | |
|---|---|
| **IDENTIFIER** | `/per/core/trajectory-vertical-diff-query-definition` |
| **STATEMENT** | The `trajectory` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple vertical levels, but not all parameters have data for the requested levels. |

| PERMISSION 31 | |
|---|---|
| IDENTIFIER | `/per/core/trajectory-custom-dimension-diff-query-definition` |
| STATEMENT | The `trajectory` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

### 7.4.7. Corridor Query

The Corridor query returns data along and around the path defined by the coords parameter. Logic for identifying the best match for the requested corridor will depend on the `Collection` and is at the discretion of the query service implementer.

Path = {root}/collections/{collectionId}/corridor

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 28 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-corridor` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `corridor` query mandatory, then: |
| A | The Service Profile *SHALL* include a requirement mandating the `corridor` query. |
| B | The `corridor` requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of `width-units` values<br>• Enumerated list of `height-units` values<br>• Enumerated list of the operations that the query supports (i.e. GET, POST) |

| PERMISSION 32 | |
|---|---|
| IDENTIFIER | `/per/core/corridor-mandatory-query-definition` |
| STATEMENT | The `corridor` requirement *MAY* specify which OGC API-EDR `corridor` query parameters are mandatory for compliance with the profile. |

## PERMISSION 33

| | |
|---|---|
| **IDENTIFIER** | `/per/core/corridor-timestep-diff-query-definition` |
| **STATEMENT** | The `corridor` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

## PERMISSION 34

| | |
|---|---|
| **IDENTIFIER** | `/per/core/corridor-vertical-diff-query-definition` |
| **STATEMENT** | The `corridor` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple vertical levels, but not all parameters have data for the requested levels. |

## PERMISSION 35

| | |
|---|---|
| **IDENTIFIER** | `/per/core/corridor-resolution-x` |
| **STATEMENT** | The `corridor` query requirement *MAY* specify rules that constrain the valid range of resolution-x values. |

## PERMISSION 36

| | |
|---|---|
| **IDENTIFIER** | `/per/core/corridor-resolution-y` |
| **STATEMENT** | The `corridor` query requirement *MAY* specify rules that constrain the valid range of resolution-y values. |

## PERMISSION 37

| | |
|---|---|
| **IDENTIFIER** | `/per/core/corridor-resolution-z` |
| **STATEMENT** | The `corridor` query requirement *MAY* specify rules that constrain the valid range of resolution-z values. |

| PERMISSION 38 | |
|---|---|
| IDENTIFIER | `/per/core/corridor-custom-dimension-diff-query-definition` |
| STATEMENT | The `corridor` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

### 7.4.8. Locations Query

The Locations query returns data for the named location. Logic for identifying the best match for the coordinate will depend on the `Collection` and is at the discretion of the query service implementer. If a location id is not defined, the API implementation instance SHALL return a GeoJSON features array of valid location identifiers, the schema of the GeoJSON response SHOULD be defined in the OpenAPI definition of the EDR service.

Path = {root}/collections/{collectionId}/locations

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 29 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-locations` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *restricts* data queries by making the `locations` query mandatory, then: |
| A | The Service Profile *SHALL* include a requirement mandating the `locations` query. |
| B | The `locations` query requirement *SHALL* specify the following:<br>• Enumerated list of `output_format` types<br>• The `default_output_format`<br>• Enumerated list of `crs_details` values<br>• Enumerated list of the operations that the query supports (i.e. GET, POST) |

**Example 1:**

| | |
|---|---|
| identifier | /per/core/locations-mandatory-query-definition |
| statement | The `locations` requirement *MAY* specify which OGC API-EDR `locations` query parameters are mandatory for compliance with the profile. |

**Example 2:**

| PERMISSION 39 | |
|---|---|
| IDENTIFIER | `/per/core/locations-mandatory-locationid-definition` |
| STATEMENT | The `locations` query requirement *MAY* specify a list of required `locationId` values. |

| PERMISSION 40 | |
|---|---|
| IDENTIFIER | `/per/core/location-timestep-diff-query-definition` |
| STATEMENT | The `location` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for multiple timesteps, but not all parameters have data for the requested timesteps. |

| PERMISSION 41 | |
|---|---|
| IDENTIFIER | `/per/core/location-custom-dimension-diff-query-definition` |
| STATEMENT | The `location` query requirement *MAY* specify rules on how the service should respond if a query is made for several parameters for a custom dimension value range, but not all parameters have data for the requested dimension values. |

### 7.4.9. Instances Query

Having multiple versions or `Instances` of the same `Collection`, where the same information is reprocessed or regenerated is not unusual. Although these versions could be described as new Collections the `Instance` query type allows this data to be described as different views of the same `Collection`.

Path = {root}/collections/{collectionId}/instances

Dependencies: OGC API — Environmental Data Retrieval Standard

| REQUIREMENT 30 | |
|---|---|
| IDENTIFIER | `/req/core/data-query-instances` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard *extends* data queries by making the `instances` within a `Collection` queryable, then: |

| REQUIREMENT 30 | |
|---|---|
| A | `instances` *SHALL* be defined in the `data_queries` enumerated list. |
| B | The Service Profile *SHALL* specify the rules that the `instanceId` values must follow. |
| C | Those rules *SHALL* include a brief description explaining how the `instanceId` is generated. |
| D | Those rules *SHALL* be specified using either:<br>• identifier string<br>• Regular expression defining valid string patterns. |
| E | The Service Profile *MAY* define a `instanceId` string which represents either a default or the latest value of `InstanceId`. |

### 7.4.9.1. Parameter queryType

Path — Instance Query {root}/collections/{collectionId}/instances/{instanceId}/{queryType}

## 7.4.10. Items Query

Paths: * {root}/collections/{collectionId}/items * {root}/collections/{collectionId}/items/{itemid}

Dependencies

- OGC API — Features — Part 1: Core
- OGC API — Environmental Data Retrieval Standard

The EDR API Items query is like an OGC API — Features Part 1: Core endpoint. The Items query offers some basic compatibility with API-Features for convenience. It may be used to catalog pre-existing EDR sampling features. The pre-existence of an EDR sampling feature may be because a particular query has been cached for later use, such as a monitoring location. Or there may be a catalog of spatio-temporal sampling features such as domains of anomalies in a dataset. A GeoJSON-compatible JSON-Schema is specified to document an EDR API query endpoint and valid query parameters including time range, parameters, and spatial characteristics. A service can define a custom GeoJSON schema in the OpenAPI definition for the service, with the default being the edr-geojson schema if no alternative is documented.

### 7.4.10.1. ItemID parameter

If an `itemId` is not specified, the query will return a list of the available `itemIds`. This behavior is specified in OGC API - Features.

#### 7.4.10.2. Limit parameter

Paging restrictions (limit parameter provided in the request, multi-page response).

| REQUIREMENT 31 | |
|---|---|
| IDENTIFIER | `/req/core/paging-support` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile of the OGC API — Environmental Data Retrieval Standard is *extended* to support paging, then: |
| A | A requirement *SHALL* be created for each combination of query pattern and output format that must support paging. |
| B | Each paging requirement *SHALL* specify the default number of items to return per page request. |

## 7.5. General Requirements

### 7.5.1. HTTP Status Codes

HTTP response

- Response status codes

| REQUIREMENT 32 | |
|---|---|
| IDENTIFIER | `/req/core/status-codes` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | A Service Profile of the OGC API — Environmental Data Retrieval Standard *SHALL* require that the definitions of all http status codes SHALL be provided. |
| A | These definitions *SHALL* provide the following:<br>• A description of the cause of the response.<br>• A JSON schema for the message body structure |

### 7.5.2. Links

- Response links

| REQUIREMENT 33 | |
|---|---|
| IDENTIFIER | `/req/core/links` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Profile of the OGC API — Environmental Data Retrieval Standard *restricts* valid responses to only those which include links, then: |
| A | The Profile *SHALL* require that link objects are included in a response. |
| B | The Profile *SHALL* define the required link objects in full. |
| C | The link objects *SHALL* require that the `href`, `rel` and `type` attributes are populated. |
| STATEMENT | for example: <br> `{ "href": "https://creativecommons.org/licenses/by-nc/4.0/", "rel": "licence", "type": "text/html" }, { "href": "https://docs.ogc.org/cs/21-069r2/21-069r2.html", "rel": "service-doc", "title": "CoverageJSON Community Standard v1.0" "type": "text/html" }` |

### 7.5.3. Asynchronous Queries

While Web protocols typically use request-response operations, there is also support for asychronous operations.

HTTP Asynchronous — This requirement address the use of HTTP asynchronous operations such as Webhooks and Callbacks.

| REQUIREMENT 34 | |
|---|---|
| IDENTIFIER | `/req/core/asynchronous` |
| INCLUDED IN | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| STATEMENT | If a Service Profile *extends* the OGC API — Environmental Data Retrieval Standard with support for asynchronous operations, then: |
| A | Requirements *SHALL* be defined for each query type that is asynchronous |
| B | Each asynchronous query type requirement *SHALL* define the HTTP Status Code and provide a message schema and text used to inform the user that the response is asynchronous. |
| C | Each asynchronous query type requirement *SHALL* document the mechanism for delivering the result of the asynchronous query. |

| PERMISSION 42 | |
|---|---|
| IDENTIFIER | `/per/core/asynchronous` |
| STATEMENT | The documentation of the mechanism for delivering the result of the asynchronous query *MAY* be provided through a link to an external document. |

### 7.5.4. EDR Part 2 Publish Subscribe Workflows

Publish-Subscribe — This requirement addresses the use of Publish-Subscribe protocols. These are protocols supported in addition to HTTP.

| REQUIREMENT 35 | |
|---|---|
| **IDENTIFIER** | `/req/core/pubsub` |
| **INCLUDED IN** | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| **STATEMENT** | If a Profile of the OGC API — Environmental Data Retrieval Standard *extends* the supported operations to include Publish-Subscribe operations, then: |
| **A** | Support for the OGC API — Environmental Data Retrieval — Part 2: Publish-Subscribe workflow Standard *SHALL* be required. |
| **B** | The pubsub requirement *SHALL* specify the channels that *SHALL* be supported |
| **C** | The pubsub requirement *SHALL* specify the payloads that a pubsub channel *SHALL* support |

8

# MEDIA TYPES FOR ANY DATA ENCODING(S)

# 8 MEDIA TYPES FOR ANY DATA ENCODING(S)

A section describing the MIME-types to be used is mandatory for any standard involving data encodings. If no suitable MIME type exists in http://www.iana.org/assignments/media-types/index.html then this section may be used to define a new MIME type for registration with IANA.

A

# ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

# A ANNEX A (INFORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE (NORMATIVE)

## A.1. Conformance Class Core

| CONFORMANCE CLASS A.1: CONFORMANCE CLASS 'CORE' | |
|---|---|
| IDENTIFIER | `http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/conf-class-core` |
| REQUIREMENTS CLASS | Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/req-class-core` |
| CONFORMANCE TESTS | Abstract test A.1: `/conf/core/modspec`<br>Abstract test A.2: `/conf/core/publishing`<br>Abstract test A.3: `/conf/core/api`<br>Abstract test A.4: `/conf/core/edr-conformant`<br>Abstract test A.5: `/conf/core/root`<br>Abstract test A.6: `/conf/core/requirements-set`<br>Abstract test A.7: `/conf/core/parameter-names`<br>Abstract test A.8: `/conf/core/collectionid`<br>Abstract test A.9: `/conf/core/extent`<br>Abstract test A.10: `/conf/core/output-format`<br>Abstract test A.11: `/conf/core/paging-support`<br>Abstract test A.12: `/conf/core/status-codes`<br>Abstract test A.13: `/conf/core/links`<br>Abstract test A.14: `/conf/core/data-query`<br>Abstract test A.17: `/conf/core/data-query-area`<br>Abstract test A.20: `/conf/core/data-query-corridor`<br>Abstract test A.18: `/conf/core/data-query-cube`<br>Abstract test A.22: `/conf/core/data-query-instances`<br>Abstract test A.21: `/conf/core/data-query-locations`<br>Abstract test A.15: `/conf/core/data-query-position`<br>Abstract test A.16: `/conf/core/data-query-radius`<br>Abstract test A.19: `/conf/core/data-query-trajectory`<br>Abstract test A.23: `/conf/core/asynchronous` |

## CONFORMANCE CLASS A.1: CONFORMANCE CLASS 'CORE'

Abstract test A.24: `/conf/core/pubsub`

## ABSTRACT TEST A.1

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/modspec` |
| **REQUIREMENT** | Requirement 1: `/req/core/modspec` |
| **TEST PURPOSE** | Validate that the service profile is compliant with the OGC Modular Specification. |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile is compliant with the OGC Modular Specification. |

## ABSTRACT TEST A.2

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/publishing` |
| **REQUIREMENT** | Requirement 11: `/req/core/publishing` |
| **TEST PURPOSE** | Validate that an OpenAPI description of the service profile is available |
| **TEST METHOD** | |
| **STEP** | Verify that an OpenAPI document describing the service profile exists<br><br>Verify that the OpenAPI document is compliant with the OpenAPI 3.1 specification<br><br>Verify that the OpenAPI document includes schemas for the `Collection` and `Instance` query responses<br><br>Verify that the `Collection` and `Instance` schemas include all of the queries defined in the data-query requirement<br><br>Verify that the `Collection` and `Instance` schema include Path objects for each of the data_queries defined in the service profile.<br><br>Verify that the Path Object define Items for the HTTP operations defined by the data query requirements in the service profile<br><br>Verify that the Path Item definitions include the enumerated lists defined by the data query requirements in the service profile |

## ABSTRACT TEST A.3

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/api` |
| **REQUIREMENT** | Requirement 1-12: `/req/core/api` |
| **TEST PURPOSE** | Validate that the service profile includes a requirement specifying the OpenAPI versions that implementations shall support. |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile includes a requirement defining the OpenAPI versions that implementations shall support.<br><br>Verify that the service profile includes a conformance class testing that OpenAPI versions are supported. |

## ABSTRACT TEST A.4

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/edr-conformant` |
| **REQUIREMENT** | Requirement 2: `/req/core/edr-conformant` |
| **TEST PURPOSE** | Validate that any implementation conforming to the Service Profile Standard demonstrates conformance with the OGC API-EDR Standard. |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile specifies that regardless of any service profile specific requirements, implementations shall function as OGC API-EDR Part 1 — Core compliant API's. |

## ABSTRACT TEST A.5

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/root` |
| **REQUIREMENT** | Requirement 4: `/req/core/root` |
| **TEST PURPOSE** | Validate that the service profile defines the service landing page |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile defines a `Title` for the service<br><br>Verify that the service profile defines the `Links` required for the service |

## ABSTRACT TEST A.5

Verify that each link defined for the service has a `href` and `rel` attribute.

Verify that any `Description` attributes in the service profile have a defined value.

Verify that any `Attribution` attributes in the service profile have a defined value.

Verify that any `Keywords` attributes in the service profile have defined values.

Verify that any `Provider` attributes in the service profile have `name` and `url` attributes.

Verify that the providers `name` and `url` attributes have defined values.

Verify that any `Contact` attributes in the service profile have an `email` attributes.

Verify that contact `email` attributes has a defined value.

## ABSTRACT TEST A.6

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/requirements-set` |
| **REQUIREMENT** | Requirement 14: `/req/core/requirements-set` |
| **TEST PURPOSE** | Validate that the service profile defines at least one requirement for a collection |
| **TEST METHOD** | |
| **STEP** | Verify that the collection contains a link to the service profile definition.<br>Verify that the service profile contains at least one requirement which applies to a collection.<br>Verify that collection requirements in the service profile clearly identify which collection a requirement applies to. |

## ABSTRACT TEST A.7

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/parameter-names` |
| **REQUIREMENT** | Requirement 3: `/req/core/parameter-names` |
| **TEST PURPOSE** | Validate that the parameter_names requirement is correctly defined |

| ABSTRACT TEST A.7 | |
|---|---|
| **TEST METHOD** | |
| **STEP** | Verify that the parameter_names requirement defines a set of parameter objects<br><br>Verify that each parameter object in the set has a unique key<br><br>Verify that each parameter object in the set has a type value of “Parameter”<br><br>Verify that each parameter object in the set has a description attribute<br><br>Verify that each parameter object in the set has a unit attribute<br><br>Verify that each unit object a label attribute<br><br>Verify that each unit object a symbol attribute<br><br>Verify that each symbol object a value attribute<br><br>Verify that each symbol object a type attribute<br><br>Verify that each parameter object in the set has an observedProperty attribute<br><br>Verify that each observedProperty object has an id attribute<br><br>Verify that each observedProperty object has a label attribute |

| ABSTRACT TEST A.8 | |
|---|---|
| **IDENTIFIER** | `/conf/core/collectionid` |
| **REQUIREMENT** | Requirement 15: `/req/core/collectionid` |
| **TEST PURPOSE** | Validate that a `collectionId` requirement is correctly defined. |
| **TEST METHOD** | |
| **STEP** | Verify that a `collectionId` requirement includes an explanation of how a `collectionId` is derived.<br><br>Verify that a `collectionId` requirement specifies either an identifier string or a Regular expression. |

## ABSTRACT TEST A.9

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/extent` |
| **REQUIREMENT** | Requirement 16: `/req/core/extent` |
| **TEST PURPOSE** | Validate the service profile extent requirements are defined correctly. |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile has a requirement specifying the spatial extent of the `Collection`.<br><br>Verify that the service profile specifies either an enumerated list of CRS values or a Regular expression definition for valid CRS values.<br><br>Verify that Temporal extent requirements specify either an enumerated list of TRS values or a Regular expression definition for valid TRS values.<br><br>Verify that Vertical extent requirements specify either an enumerated list of VRS values or a Regular expression definition for valid VRS values.<br><br>Verify that Custom extent requirements specify the id of the custom dimension.<br><br>Verify that Custom extent requirements specify the custom dimension reference value.<br><br>Verify that Custom extent requirements specify the custom dimension interval value.<br><br>Verify that Custom extent requirements specify an enumerated list of the valid custom dimension values. |

## ABSTRACT TEST A.10

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/output-format` |
| **REQUIREMENT** | Requirement 22: `/req/core/output-format` |
| **TEST PURPOSE** | Validate that the service profile correctly defines output-formats |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile contains references to the schemas or format descriptions for all output-formats defined in the data queries requirements |

## ABSTRACT TEST A.11

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/paging-support` |
| **REQUIREMENT** | Requirement 31: `/req/core/paging-support` |
| **TEST PURPOSE** | Validate that paging support requirements are correctly defined |
| **TEST METHOD** | |
| **STEP** | Verify that a paging support requirement defines which query types will support output paging<br><br>Verify that a paging support requirement identifies which output formats support paging in paging-capable queries<br><br>Verify that a paging support requirement defines the default number of items to return per page request |

## ABSTRACT TEST A.12

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/status-codes` |
| **REQUIREMENT** | Requirement 32: `/req/core/status-codes` |
| **TEST PURPOSE** | Validate that the service profile defines any HTTP status code responses required by a service. |
| **TEST METHOD** | |
| **STEP** | Verify that the service profile contains HTTP status code definitions<br><br>Verify that any HTTP status code definition has a description value.<br><br>Verify that each HTTP status code definition includes a JSON schema for the response body. |

## ABSTRACT TEST A.13

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/links` |
| **REQUIREMENT** | Requirement 33: `/req/core/links` |
| **TEST PURPOSE** | Validate that link requirements are correctly defined |

## ABSTRACT TEST A.13

| | |
|---|---|
| **TEST METHOD** | |
| **STEP** | Verify that link requirements define a `href` value<br><br>Verify that link requirements define a `rel` value<br><br>Verify that link requirements define a `type` value |

## ABSTRACT TEST A.14

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query` |
| **REQUIREMENT** | Requirement 21: `/req/core/data-query` |
| **TEST PURPOSE** | Validate that a service profile correctly defines the query types that a service shall support. |
| **TEST METHOD** | |
| **STEP** | Verify that the profile defines an enumerated list of the queries that a service shall support.<br><br>Verify that for each definition in the enumerated list there is a corresponding data query requirement in the profile. |

## ABSTRACT TEST A.15

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-position` |
| **REQUIREMENT** | Requirement 23: `/req/core/data-query-position` |
| **TEST PURPOSE** | Verify that Position query requirements are defined correctly in a service profile. |
| **TEST METHOD** | |
| **STEP** | Verify a Position query requirement defines the logic in used selecting the appropriate data response.<br><br>Verify a Position query requirement defines an enumerated list of output_format types.<br><br>Verify a Position query requirement defines the default output_format.<br><br>Verify a Position query requirement defines an enumerated list of crs_details values. |

## ABSTRACT TEST A.15

| | |
|---|---|
| | Verify a Position query requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.16

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|---|---|
| **IDENTIFIER** | `/conf/core/data-query-radius` |
| **REQUIREMENT** | Requirement 24: `/req/core/data-query-radius` |
| **TEST PURPOSE** | Verify that Radius query requirements are defined correctly in a service profile. |
| **TEST METHOD** | |
| **STEP** | Verify a Radius query requirement defines an enumerated list of output_format types.<br>Verify a Radius query requirement defines the default output_format.<br>Verify a Radius query requirement defines an enumerated list of crs_details values.<br>Verify a Radius query requirement defines an enumerated list of within_units values.<br>Verify a Radius query requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.17

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-area` |
| **REQUIREMENT** | Requirement 25: `/req/core/data-query-area` |
| **TEST PURPOSE** | Verify that Area query requirements are defined correctly in a service profile. |
| **TEST METHOD** | |
| **STEP** | Verify an Area query requirement defines an enumerated list of output_format types.<br>Verify an Area query requirement defines the default output_format.<br>Verify an Area query requirement defines an enumerated list of crs_details values.<br>Verify an Area query requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.18

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-cube` |
| **REQUIREMENT** | Requirement 26: `/req/core/data-query-cube` |
| **TEST PURPOSE** | Verify that Cube query requirements are defined correctly in a service profile. |
| **TEST METHOD** | |
| **STEP** | Verify a Cube query requirement defines an enumerated list of output_format types.<br>Verify a Cube query requirement defines the default output_format.<br>Verify a Cube query requirement defines an enumerated list of crs_details values.<br>Verify a Cube query requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.19

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-trajectory` |
| **REQUIREMENT** | Requirement 27: `/req/core/data-query-trajectory` |
| **TEST PURPOSE** | Verify that Trajectory query requirements are defined correctly in a service profile. |
| **TEST METHOD** | |
| **STEP** | Verify a Trajectory query requirement defines an enumerated list of output_format types.<br>Verify a Trajectory query requirement defines the default output_format.<br>Verify a Trajectory query requirement defines an enumerated list of crs_details values.<br>Verify a Trajectory query requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.20

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-corridor` |
| **REQUIREMENT** | Requirement 28: `/req/core/data-query-corridor` |
| **TEST PURPOSE** | Verify that Corridor query requirements are defined correctly in a service profile. |

## ABSTRACT TEST A.20

| | |
|---|---|
| **TEST METHOD** | |
| **STEP** | Verify that a Corridor requirement defines an enumerated list of output_format types.<br>Verify that a Corridor requirement defines the default output_format.<br>Verify that a Corridor requirement defines an enumerated list of crs_details values.<br>Verify that a Corridor requirement defines an enumerated list of width-units values.<br>Verify that a Corridor requirement defines an enumerated list of height-units values.<br>Verify that a Corridor requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.21

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-locations` |
| **REQUIREMENT** | Requirement 29: `/req/core/data-query-locations` |
| **TEST PURPOSE** | Verify that Locations query requirements are defined correctly in a service profile. |
| **TEST METHOD** | |
| **STEP** | Verify a Locations query requirement defines an enumerated list of output_format types.<br>Verify a Locations query requirement defines the default output_format.<br>Verify a Locations query requirement defines an enumerated list of crs_details values.<br>Verify a Locations query requirement defines a list of supported HTTP operations. |

## ABSTRACT TEST A.22

| | |
|---|---|
| **IDENTIFIER** | `/conf/core/data-query-instances` |
| **REQUIREMENT** | Requirement 30: `/req/core/data-query-instances` |
| **TEST PURPOSE** | Validate than a service profile correctly defines support for `instances` |
| **TEST METHOD** | |

## ABSTRACT TEST A.22

| | |
|---|---|
| STEP | Verify that the data queries requirement enumerated list includes an entry for `instances`.<br><br>Verify that an `instanceId` requirement includes an explanation of how an `instanceId` is derived.<br><br>Verify that an `instanceId` requirement contains either an identifier string or a Regular expression rule for defining the `instanceId`. |

## ABSTRACT TEST A.23

| | |
|---|---|
| IDENTIFIER | `/conf/core/asynchronous` |
| REQUIREMENT | Requirement 34: `/req/core/asynchronous` |
| TEST PURPOSE | Validate that an asynchronous query support requirement in correctly defined |
| TEST METHOD | |
| STEP | Verify that all asynchronous query requirements in a service profile define a HTTP Status Code for the asynchronous response.<br><br>Verify that all asynchronous query requirements in a service profile define a message schema for the asynchronous response.<br><br>Verify that all asynchronous query requirements in a service profile define the text for the asynchronous response.<br><br>Verify that all asynchronous query requirements in a service profile define the mechanism for delivering the asynchronous query result. |

## ABSTRACT TEST A.24

| | |
|---|---|
| IDENTIFIER | `/conf/core/pubsub` |
| REQUIREMENT | Requirement 35: `/req/core/pubsub` |
| TEST PURPOSE | Validate Pub/Sub requirements |
| TEST METHOD | |
| STEP | Verify that the service profile specifies the OGC API-EDR Part 2 conformance class |

## ABSTRACT TEST A.24

Verify that the service profile defines the channels services shall implement.

Verify that the service profile defines the message payloads required for each of the channels in the service.

Verify the service profile contains an AsyncAPI definition that describes the channels and their messages.